INCH-POUND

MIL-PRF-1/1647B
7 October 1996
SUPERSEDING
MIL-E-1/1647A(EC)
26 November 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER
TYPE 8767

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-PRF-1.

DESCRIPTION: Geiger - Mueller, very high beta sensitivity, very thin mica window frisker probe, low background, halogen, self-quenching, ruggedized
See figures 1, 2, 3, 4, and 5
Mounting position: Any
Weight: 6.5 ounces (184.3 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ebb V dc	TA °C	Rp (external resistance) Meg	Window thickness mgs/cm^2
Maximum:	950	+55	---	2.0
Minimum:	850	-20	3.3	1.4
Test conditions:	900	---	3.3	---

GENERAL: Qualification - Required
Marking 1/
Holding period (MIL-STD-1311) 2/
Packaging 3/

1/ Each tube shall have an individual serial number which shall be legibly punched or engraved into the cathode.

2/ The tube shall meet the initial acceptance requirements for background, contamination, and photosensitivity and starting voltage, and shall not be greater than 10 volts more than the starting voltage at the beginning of the last 60-day period.

3/ Tubes shipped under Government contract must be packaged in hermetically sealed containers to withstand low external pressure which might occur in air shipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5960

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TABLE I. Group A inspection.

Inspection	Method	Conditions	Acceptance level	Symbol	Limits min	Limits max	Unit
Quality conformance inspection, part 1							
Background, contamination, and photosensitivity 2/, 3/	6201	t = 5 minutes	0.65	N/T	----	33.0	Npm
Starting voltage	6211	Pulse amplitude = 0.25 V use test circuit shown on figure 5; Nps = 200 (max)	0.65	Es	----	750	Vdc
Relative plateau slope 1/ 2/	6216	Nps = 5,000	0.65	----	----	----	----
Response count rate and current (beta) 2/ 5/	6221	t = 2 minutes	0.65	----	----	----	v----
Dead time	----	Nps = 5,000; use test circuit shown on figure 5	0.65	td	----	100	μs
Quality conformance inspection, part 2							
Shock 4/ 7	1041	Hammer angle = 20°	----	----	----	----	----
Shock, specified 4/ 7/ 9/	1042	15G, 11 ms duration, half-sine waveform; 3.4 ft/sec velocity	----	----	----	----	----
Pulse amplitude	6226	Use test circuit shown on figure 5	----	eo	13	100	v
Quality conformance inspection, part 3							
Life test (1) 1/	----	Group D; Npm = 200 (min); t = 500 hours	----	----	----	----	----
Life test endpoints							
Background, contamination, and photosensitivity	6201		----	N/T	----	60	Npm
Starting voltage	6211		----	△Es	----	±15	Vdc
Pulse amplitude	6226		----	eo	13	----	V
Variable frequency 4/ 7/ vibration	1031		----	----	----	----	----

See footnotes at end of table.

TABLE I. Group A inspection - Co ntinued.

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Inspection	Method	Conditions	Acceptance level	Symbol	Limits		Unit
					min	max	
Quality conformance inspection, part 3 Continued							
Leakage current 8 /	6205	Ebb = 500 V dc	----	Lib	----	0.5	μA dc
Temperature cycling 6/	----		----	----	----	----	----

1/ Npm - N_{850} = ±0.05 Npm maximum
Npm - N_{950} = ±0.05 Npm maximum
Npm - 5,000 cpm nominal at V dc = 900 V dc.
N_{850} = Count rate at V dc = 850 V.
N_{950} = Count rate at V dc = 950 V.

2/ Tube count rates shall be determined using a scaler having a maximum pulse pair resolving time of 10μS and a pulse height discrimination level of 0.25 volt.

3/ The tube end window shall be exposed to radiation from a General Electric 15-watt germicidal lamp and a General Electric 15-watt fluorescent lamp, or equivalents. Light sources shall be located not more than 1 inch from the mica window surface (see figure 2).

4/ The criterion for passing this test shall be compliance of a lease 80 percent of the tubes, after test, with the requirements for starting voltage and relative plateau slope.

5/ The counting rate shall be not less than 10 percent below the value of the count rate specified in the EON TC-9 excitation unit, or equivalent (see figure 3).

6/ With the tube in a field giving 100 ±10 Nps at 900 V dc, determine tube response (count rate) a each of the following temperature and in the order shown:

(1) Room temperature
(2) -20°C
(3) Room temperature
(4) +55°C
(5) Room temperature

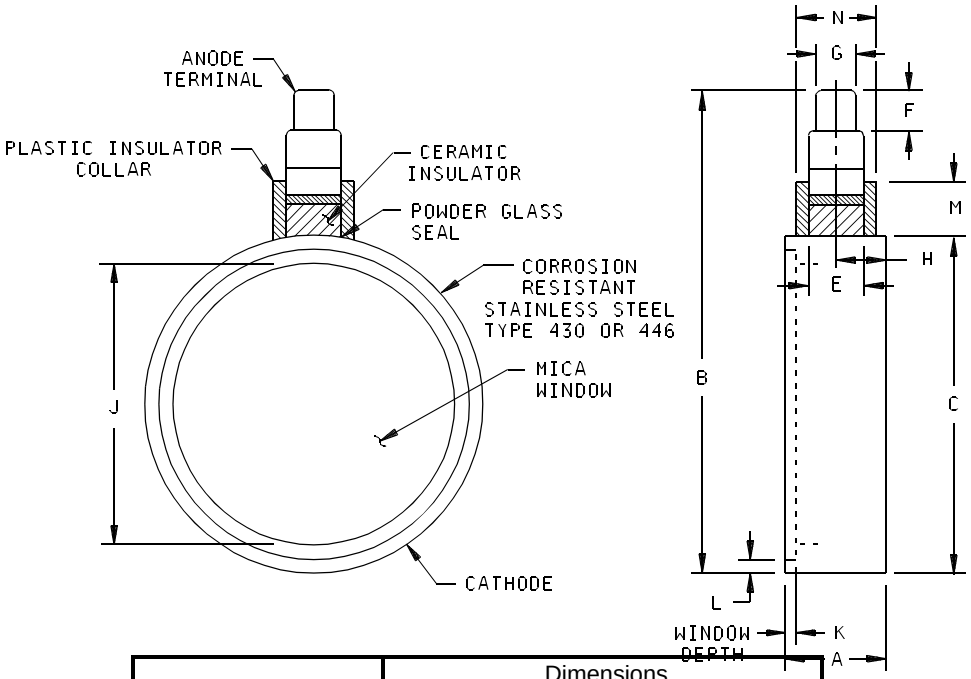
A minimum stabilization time of 30 minutes shall be allowed at each temperature. The absolute count rate at 900 V dc shall not differ from the initial readings at room temperature by more than 10 percent at any of the four subsequent temperatures.

7/ The tube shall be mounted by means of a rigid fixture which clamps the detector at the cathode as shown on figure 4.

8/ Test shall be made at ambient conditions of normal room temperature and humidity.

9/ This is an alternate test to method 1041. Either method 1041 or method 1042 may be used but only one of these tests shall be performed.

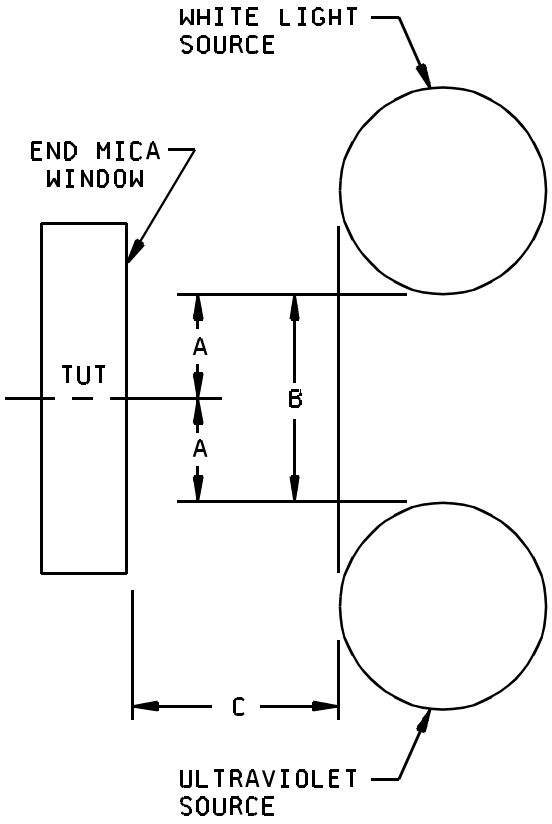
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LTR	Dimensions			
	Millimeters		Inches	
QUALITY CONFORMANCE INSPECTION, PART 2				
	Min	Max	Min	Max
A	15.09	15.85	0.594	0.624
B	------	76.20	-----	3.000
C	53.34	53.85	2.100	2.120
E	8.48	8.99	0.334	0.354
G	6.22	6.48	0.245	0.255
J	------	4.45	-----	1.750
L	1.78	2.16	0.070	0.085
M	8.43	8.76	0.332	0.345
N	12.19	12.83	0.480	0.505
REFERENCE DIMENSIONS				
F	6.35		0.250	
H	7.95		0.313	
K	1.60		0.063	

FIGURE 1. Outline drawing of electron tube type 8767.

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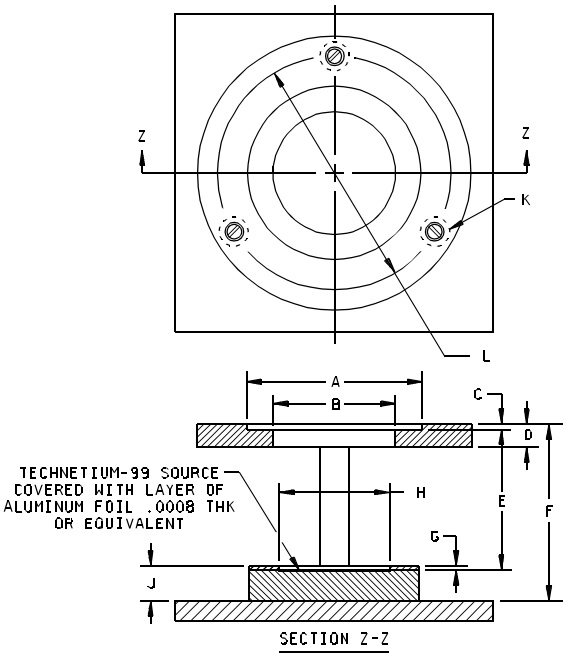


Ltr	Dimensions			
	Millimeters		Inches	
	Minimum	Maximum	Minimum	Maximum
A	12.7 Nominal		0.50 Nominal	
B	-----	25.4	-----	1.0
C	-----	25.4	-----	1.0

NOTE: Axis of tube and centers of lamps lie in plane of paper

FIGURE 2. Position of tube for photosensitivity test.

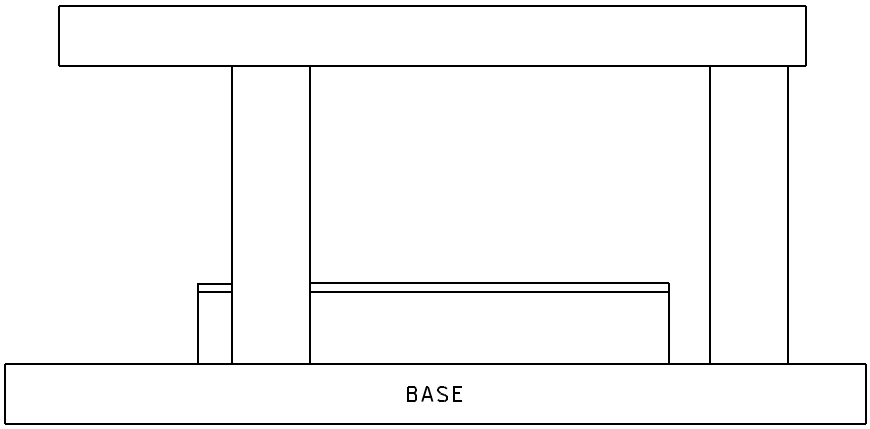
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Ltr	Dimensions				Ltr	Dimensions			
	Millimeters		Inches			Millimeters		Inches	
	Minimum	Maximum	Minimum	Maximum		Minimum	Maximum	Minimum	Maximum
A	53.721	53.848	2.115	2.120	B	44.831 Nominal		1.765 Nominal	
C	2.362 Nominal		0.093 Nominal		D	9.525 Nominal		0.375 Nominal	
E	47.625 Nominal		1.875 Nominal		F	54.102	66.802	2.13	2.63
G	3.962 Nominal		0.156 Nominal		H	49.200 Nominal		1.937 Nominal	
J	14.732 Nominal		0.580 Nominal		L	103.124 Nominal		4.060 Nominal	
K	Post, 3 each, 0.50 diameter, equally spaced								

FIGURE 3. Excitation fixture.

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Beta fixture:

Bogey value for beta response is 7,000 Npm.

FIGURE 3. Excitation fixture - Continued.

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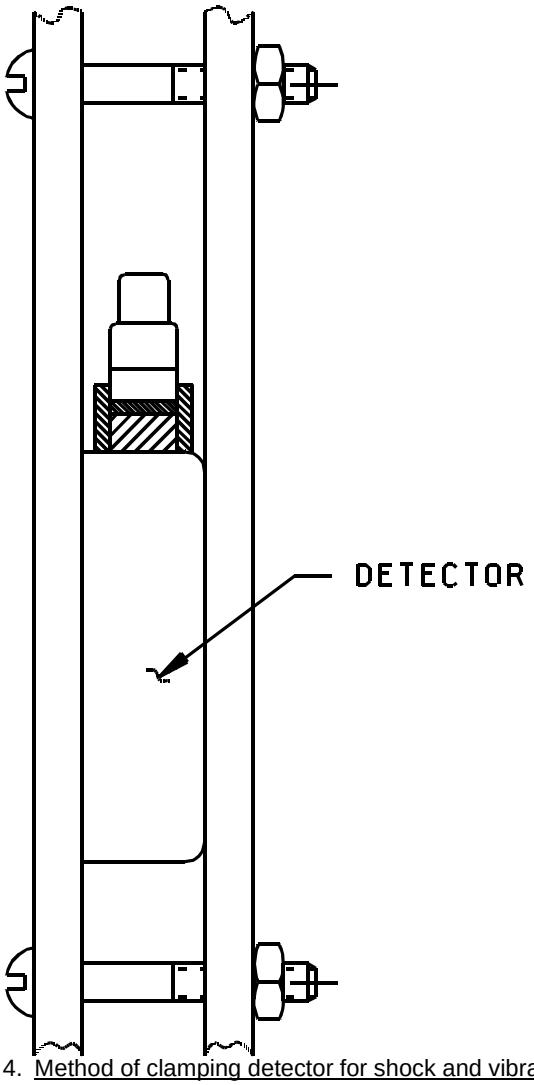


FIGURE 4. Method of clamping detector for shock and vibration test.

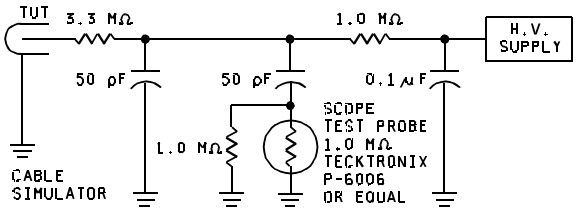


FIGURE 5. Test Circuit.

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CONCLUDING MATERIAL

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Preparing activity:
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3. DOCUMENT TITLE
ELECTRON TUBE, RADIATION COUNTER TYPE 8767

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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